

AI-Powered Customer Support Automation Platform

Capstone Project — Summer School '26 (n8n)

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1. Problem Analysis

Business Context

A growing software company receives customer support requests through multiple channels — email, website forms, and chat. As the customer base scales, manually categorizing tickets, assigning them to the right agents, tracking resolution, and following up with customers becomes slow and error-prone. This leads to delayed responses, inconsistent prioritization, duplicate effort, and declining customer satisfaction.

This project automates the full support ticket lifecycle — from intake to resolution to feedback — using a modular, AI-powered n8n platform.

Stakeholders

Stakeholder	Interest
Customers	Fast acknowledgment, timely resolution, clear communication
Support Agents	Clear ticket ownership, less manual triage
Support Managers	Visibility into critical issues, workload distribution, escalation control
Company Leadership	Customer satisfaction metrics, operational efficiency, cost reduction

Pain Points

- Manual ticket categorization is slow and inconsistent across agents
- No automatic prioritization — critical issues can sit unnoticed alongside minor ones
- No structured escalation path for urgent issues
- Customers aren't proactively updated on ticket status
- No systematic feedback collection or reporting on support performance

Objectives

- Automatically capture and log every incoming ticket
- Use AI to classify ticket category and priority without human intervention
- Route tickets to the correct team automatically
- Escalate critical (P1) tickets to a human for immediate review and approval
- Keep customers informed at every stage (acknowledgment → assignment → resolution)
- Collect structured feedback after resolution
- Maintain a full audit trail and generate analytics for management

2. Workflow Architecture

Overall System Architecture

The platform consists of 5 independent, interconnected n8n workflows, each responsible for one stage of the ticket lifecycle. Workflows communicate using n8n's Execute Sub-workflow trigger/action pattern, passing structured JSON data forward at each stage. A Google Sheet acts as the central ticket database (Support Tickets DB), with a second tab (Feedback) for survey responses.

Workflow Interaction Diagram

See the accompanying architecture diagram (02_Architecture_Diagram) for the full visual flow. Summary of stages:

Customer	→	[WF1: Ticket Collection & Registration]	
	→	[WF2: AI Classification & Prioritization]	(Gemini)
	→	[WF3: Agent Assignment & Escalation]	(Switch + IF)
	→	[WF4: Customer Communication & Resolution]	(Webhook: resolve)
	→	[WF5: Feedback Collection & Analytics]	(Webhook: feedback, Cron: report)

Event Flow Between Workflows

- Webhook (POST /ticket-intake) → Workflow 1 starts execution
- Workflow 1 completes ticket registration → calls Workflow 2 via Execute Sub-workflow, passing ticket fields
- Workflow 2 calls Gemini AI to classify category/priority → updates the sheet → calls Workflow 3
- Workflow 3 routes the ticket (Switch node on category) → checks priority (IF node) → escalates via email if P1 → calls Workflow 4
- Workflow 4 sends the assignment update email immediately. Later, an independent webhook (POST /ticket-resolve) — representing a real agent action — triggers the resolution branch, which calls Workflow 5
- Workflow 5 sends a feedback survey. A separate webhook (POST /submit-feedback) logs customer responses. A Cron-scheduled branch independently reads all tickets and emails a summary analytics report

3. Implementation Summary

Workflow	Node Count (approx.)	Trigger Type
1. Ticket Collection & Registration	5	Webhook
2. AI Ticket Classification & Prioritization	4	Execute Sub-workflow Trigger
3. Agent Assignment & Escalation	11	Execute Sub-workflow Trigger
4. Customer Communication & Resolution Tracking	7	Execute Sub-workflow Trigger + Webhook
5. Feedback Collection & Support Analytics	8	Execute Sub-workflow Trigger + Webhook + Schedule (Cron)

Total: ~35 nodes across 5 workflows — exceeding the minimum requirement of 4–6 workflows and 20–25 nodes.

4. Advanced Features Implemented

Feature	Where Implemented
AI-powered decision making	Workflow 2 — Google Gemini classifies ticket category and priority from natural language
Human approval step	Workflow 3 — P1 (critical) tickets trigger an escalation email requiring manager review before proceeding
Error handling	Resolved data-loss issues across Gmail response nodes; used explicit node references (\$('NodeName').item.json) to ensure reliable data flow past nodes that overwrite payload data
Logging and audit trail	Every stage of the ticket lifecycle is logged to Google Sheets (status, category, priority, assigned_team, timestamps)
Scheduled workflow (Cron)	Workflow 5 — periodic analytics report generation and email
Webhook-triggered workflows	Workflow 1 (ticket intake), Workflow 4 (resolution update), Workflow 5 (feedback submission)
Conditional branching	Workflow 3 — Switch node (category routing) + IF node (priority-based escalation)

5. Tech Stack

- Automation platform: n8n (self-hosted via Docker, WSL2 / Ubuntu 24.04)
- AI model: Google Gemini (via Google AI Studio API)
- Database: Google Sheets (Support Tickets DB — Sheet1 + Feedback tab)
- Notifications: Gmail (OAuth2)
- Trigger types: Webhook (REST), Execute Sub-workflow, Schedule (Cron)